## Exercise 5.1 SOLUTION

**Resolve the following into partial fractions:**

**Q1.** $\frac{2}{x^2-1}$

**Sol:** Let

$$\frac{2}{x^2-1}=\frac{2}{(x-1)(x+1)}=\frac{A}{x-1}+\frac{B}{x+1} \quad \text{_____ (i)}$$

Multiply both sides by $(x-1)(x+1)$ we get

$$2=A(x+1)+B(x-1) \quad \text{_____ (ii)}$$

Put $x-1=0 \Rightarrow x=1$ in (ii)

$$2=A(1+1)+B(1-1) \Rightarrow 2=2A \Rightarrow A=1$$

Now put $x+1=0 \Rightarrow x=-1$ in (ii)

$$2=A(-1+1)+B(-1-1)$$

$$2=A(0)+B(-2)$$

$$2=B(-2) \Rightarrow B=-1$$

Put values of A and B in (i)

Hence $\frac{2}{x^2-1}=\frac{1}{x-1}-\frac{1}{x+1}$

نوٹ:

اگر denominator میں factors نہ ہوں تو پہلے factors بناتے ہیں پھر اس کی type کا اندازہ ہو گا۔

**Q2.** $\frac{a-b}{(x-a)(x-b)}, a \neq b$

**Sol:** Let $\frac{a-b}{(x-a)(x-b)}=\frac{A}{x-a}+\frac{B}{x-b}$ _____ (i)

Multiply both sides by $(x-a)(x-b)$ we get

$$a-b=A(x-b)+B(x-a) \quad \text{_____ (ii)}$$

Put $x-a=0 \Rightarrow x=a$ in (ii)

$$a-b=A(a-b)+B(a-a)$$

$$a-b=A(a-b)+0$$

$$a-b=A(a-b) \Rightarrow A=\frac{\cancel{a-b}}{\cancel{a-b}} \Rightarrow A=1$$

Put $x-b=0 \Rightarrow x=b$ in (ii)

$$a-b=A(b-b)+B(b-a)$$

$$a-b=A(0)+B(b-a)$$

$$a-b=-B(a-b) \Rightarrow B=\frac{-\cancel{(a-b)}}{\cancel{(a-b)}} \Rightarrow B=-1$$

Put values of A and B in (i)

Hence $\frac{a-b}{(x-a)(x-b)}=\frac{1}{x-a}+\frac{-1}{x-b}=\frac{1}{x-a}-\frac{1}{x-b}$

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Q3. $\dfrac{x^2+1}{(x+1)(x-1)}$

Sol: $\dfrac{x^2+1}{(x+1)(x-1)} = \dfrac{x^2+1}{x^2-1}$ (Improper fraction)

$$\begin{array}{r} 1 \\ x^2-1 \overline{)\ x^2+1} \\ \pm x^2 \mp 1 \\ \hline 2 \end{array}$$

$\dfrac{x^2+1}{x^2-1} = 1 + \dfrac{2}{x^2-1}$ _____ (i)

Now take $\dfrac{2}{x^2-1} = \dfrac{2}{(x-1)(x+1)}$

Let $\dfrac{2}{(x-1)(x+1)} = \dfrac{A}{x-1} + \dfrac{B}{x+1}$ _____ (ii)

Multiply both sides by $(x-1)(x+1)$ we get

$2 = A(x+1) + B(x-1)$ _____ (iii)

Put $x-1=0 \Rightarrow x=1$ in (iii)

$2 = A(1+1) + B(1-1)$

$2 = A(2) + 0 \Rightarrow A = 1$

Put $x+1=0 \Rightarrow x=-1$ in (iii)

$2 = A(-1+1) + B(-1-1)$

$2 = A(0) + B(-2) \Rightarrow 2 = -2B \Rightarrow B = -1$

Put values in (ii)

$$\frac{2}{(x-1)(x+1)} = \frac{1}{x-1} + \frac{-1}{x+1}$$

(i) become

Hence $\dfrac{x^2+1}{(x-1)(x+1)} = 1 + \dfrac{1}{x-1} - \dfrac{1}{x+1}$

Q4. $\dfrac{2x+3}{(x+1)(x+2)(x+3)}$

Sol: $\dfrac{2x+3}{(x+1)(x+2)(x+3)}$

Suppose

$\dfrac{2x+3}{(x+1)(x+2)(x+3)} = \dfrac{A}{x+1} + \dfrac{B}{x+2} + \dfrac{C}{x+3}$ _____ (i)

Multiply both sides by $(x+1)(x+2)(x+3)$ we get

$2x+3 = A(x+2)(x+3) + B(x+1)(x+3) + C(x+1)(x+2)$ _____ (ii)

Put $x+1=0 \Rightarrow x=-1$ in (ii)

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$$2(-1)+3=A(-1+2)(-1+3)+B(-1+1)(-1+3)+C(-1+1)(-1+2)$$

$$-2+3=A(1)(2)+B(0)+C(0)$$

$$1=\ 2A\Rightarrow A=\frac{1}{2}$$

Put $x+2=0\Rightarrow x=-2$ in (ii)

$$2(-2)+3=A(-2+2)(-2+3)+B(-2+1)(-2+3)+C(-2+1)(-2+2)$$

$$-4+3=A(0)+B(-1)(1)+C(0)$$

$$-1=-1B\Rightarrow B=1$$

Put $x+3=0\Rightarrow x=-3$ in (ii)

$$2(-3)+3=A(-3+2)(-3+3)+B(-3+1)(-3+3)+C(-3+1)(-3+2)$$

$$-6+3=A(0)+B(0)+C(-2)(-1)$$

$$-3=2C\Rightarrow\ -3/2=C$$

Equation (I) become

$$\frac{2x+3}{(x+1)(x+2)(x+3)}=\frac{1}{2(x+1)}+\frac{1}{(x+2)}+\frac{-3/2}{(x+3)}$$

Hence $\dfrac{2x+3}{(x+1)(x+2)(x+3)}=\dfrac{1}{2(x+1)}+\dfrac{1}{(x+2)}-\dfrac{3}{2(x+3)}$

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**Q5.** $\dfrac{x^2+4x+5}{(x+1)(x^2+5x+6)}$

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**Sol:**

$$\frac{x^2+4x+5}{(x+1)(x^2+5x+6)}=\frac{x^2+4x+5}{(x+1)(x^2+3x+2x+6)}$$

$$\frac{x^2+4x+5}{(x+1)(x^2+5x+6)}=\frac{x^2+4x+5}{(x+1)x(x+3)+2(x+3)}=\frac{x^2+4x+5}{(x+1)(x+2)(x+3)}$$

Suppose

$$\frac{x^2+4x+5}{(x+1)(x+2)(x+3)}=\frac{A}{x+1}+\frac{B}{x+2}+\frac{C}{x+3}\quad\text{_____ (i)}$$

Multiply by $(x+1)(x+2)(x+3)$ both sides

$$x^2+4x+5=A(x+2)(x+3)+B(x+1)(x+3)+C(x+1)(x+2)\quad\text{_____ (ii)}$$

Put $x+1=0\Rightarrow x=-1$ in (ii)

$$(-1)^2+4(-1)+5=A(-1+2)(-1+3)+B(-1+1)(-1+3)+C(-1+1)(-1+2)$$

$$1-4+5=A(1)(2)+B(0)+C(0)$$

$$2=\ 2A\Rightarrow A=1$$

put $x+2=0\Rightarrow x=-2$ in (ii)

$$(-2)^2+4(-2)+5=A(-2+2)(-2+3)+B(-2+1)(-2+3)+C(-2+1)(-2+2)$$

$$4-8+5=A(0)+B(-1)(1)+C(0)$$

$$1=\ -1B\Rightarrow B=-1$$

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Put $x+3=0 \Rightarrow x=-3$ in (ii)

$(-3)^2+4(-3)+5=A(-3+2)(-3+3)+B(-3+1)(-3+3)+C(-3+1)(-3+2)$

$9-12+5=A(0)+B(0)+C(-2)(-1)$

$2 = (2)C \Rightarrow C=1$

I Become

Hence $\dfrac{x^2+4x+5}{(x+1)(x^2+5x+6)}=\dfrac{1}{x+1}-\dfrac{1}{x+2}+\dfrac{1}{x+3}$

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**Q6.** $\dfrac{4x^3+5x^2-3x-2}{x^2-1}$

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**Sol:** $\dfrac{4x^3+5x^2-3x-2}{x^2-1}$ (improper fraction)

$$
\begin{array}{r|l}
x^2-1 & 4x+5 \\
\cancel{4x^3}+5x^2-3x-2 & \\
\cancel{\mp 4x^3} \quad \mp 4x & \\
\hline
\cancel{5x^2}+x-2 & \\
\cancel{\pm 5x^2} \quad \mp 5 & \\
\hline
x+3 &
\end{array}
$$

$\dfrac{4x^3+5x^2-3x-2}{x^2-1}=4x+5+\dfrac{x+3}{x^2-1} \rightarrow (i)$

Now take $\dfrac{x+3}{x^2-1}=\dfrac{x+3}{(x+1)(x-1)}$

Suppose $\dfrac{x+3}{(x-1)(x+1)}=\dfrac{A}{x-1}+\dfrac{B}{x+1} \rightarrow (ii)$

Multiply by $(x-1)(x+1)$

$x+3=A(x+1)+B(x-1) \rightarrow (iii)$

Put $x-1=0 \Rightarrow x=1$ in eq. (iii)

$1+3=A(1+1)+B(1-1)$

$4=A(2) \Rightarrow \boxed{A=2}$

Put $x+1=0 \Rightarrow x=-1$ in (iii)

$-1+3=A(-1+1)+B(-1-1)$

$2=A(0)+B(-2) \Rightarrow 2=-2B \Rightarrow \boxed{B=-1}$

Eq. (ii) Become

$\dfrac{x+3}{x^2-1}=\dfrac{2}{x-1}+\dfrac{-1}{x+1}$

Hence $\dfrac{4x^3+5x^2-3x-2}{x^2-1}=4x+5+\dfrac{2}{x-1}-\dfrac{1}{x+1}$ (equation (i) become)

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**Q7.** $\dfrac{3x^2-12x+11}{(x-1)(x-2)(x-3)}$

**Sol:** Suppose $\dfrac{3x^2-12x+11}{(x-1)(x-2)(x-3)}=\dfrac{A}{(x-1)}+\dfrac{B}{(x-2)}+\dfrac{C}{x-3}\rightarrow(i)$

Multiply by $(x-1)(x-2)(x-3)$ on both sides

$3x^2-12x+11=A(x-2)(x-3)+B(x-1)(x-3)+C(x-1)(x-2)\rightarrow II$

Put $x-1=0\Rightarrow x=1$ in eq. (ii)

$3(1)^2-12(1)+11=A(1-2)(1-3)+B(1-1)(1-3)+C(1-1)(1-2)$

$3-12+11=A(-1)(-2)+B(0)+C(0)$

$2=2A\Rightarrow \boxed{A=1}$

Put $x-2=0\Rightarrow x=2$ in (ii)

$3(2)^2-12(2)+11=A(2-2)(2-3)+B(2-1)(2-3)+C(2-1)(2-2)$

$12-24+11=A(0)+B(1)(-1)+C(0)$

$-1=-B\Rightarrow \boxed{B=1}$

Put $x-3=0\Rightarrow x=3$ in (ii)

$3(3)^2-12(3)+11=A(3-2)(3-3)+B(3-1)(3-3)+C(3-1)(3-2)$

$27-36+11=A(0)+B(0)+C(2)(1)$

$2=2C\Rightarrow \boxed{C=1}$

Put values of $A, B$ and $C$ in eq. (i)

Hence $\dfrac{3x^2-12x+11}{(x-1)(x-2)(x-3)}=\dfrac{1}{x-1}+\dfrac{1}{x-2}+\dfrac{1}{x-3}$

**Q8.** $\dfrac{(x-1)(x-2)(x-3)}{(x-4)(x-5)(x-6)}$

**Sol:** $\dfrac{(x-1)(x-2)(x-3)}{(x-4)(x-5)(x-6)}=\dfrac{(x-1)(x^2-3x-2x+6)}{(x-4)(x^2-6x-5x+30)}$

$\dfrac{(x-1)(x^2-5x+6)}{(x-4)(x^2-11x+30)}=\dfrac{x^3-5x^2+6x-x^2+5x-6}{x^3-11x^2+30x-4x^2+44x-120}=\dfrac{x^3-6x^2+11x-6}{x^3-15x^2+74x-120}$

$=1+\dfrac{9x^2-63x+114}{x^3-15x^2+74x-120}=1+\dfrac{9x^2-63x+114}{(x-4)(x-5)(x-6)}\rightarrow(i)$

$$\begin{array}{r|l} x^3-15x^2+74x-120 & \overset{1}{\cancel{x^3}}-6x^2+11x-6 \\ & \cancel{x^3}-15x^2\pm74x-120 \\ & \quad -\quad +\qquad\quad + \\ \hline & 9x^2-63x+114 \end{array}$$

Now we take

Let $\dfrac{9x^2-63x+114}{(x-4)(x-5)(x-6)}=\dfrac{A}{(x-4)}+\dfrac{B}{(x-5)}+\dfrac{C}{(x-6)}\rightarrow(ii)$

Multiply by both sides $(x-4)(x-5)(x-6)$ we get

$9x^2-63x+114=A(x-5)(x-6)+B(x-4)(x-6)+C(x-4)(x-5)\rightarrow(iii)$

Put $x-4=0\Rightarrow x=4$ in eq. (iii)

$9(4)^2-63(4)+114=A(4-5)(4-6)+B(0)+C(0)$

$144-252+114=A(-1)(-2)$

$6=2A\Rightarrow \boxed{A=3}$

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Put $x-5=0 \Rightarrow x=5$ in Eq. (iii)

$9(5)^2-63(5)+114=A(0)+B(5-4)(5-6)+C(0)$

$225-315+114=B(1)(-1)$

$24=-B \Rightarrow \boxed{B=-24}$

Put $x-6=0 \Rightarrow x=6$ in Eq. (iii)

$9(6)^2-63(6)+114=A(0)+B(0)+C(6-4)(6-5)$

$324-378+114=0+0+C(2)(1)$

$60=2C \Rightarrow \boxed{C=30}$

Eq. (i) Become

$$\frac{(x-1)(x-2)(x-3)}{(x-4)(x-5)(x-6)}=1+\frac{3}{(x-4)}+\frac{-24}{x-5}+\frac{30}{x-6}=1+\frac{3}{x-4}-\frac{24}{x-5}+\frac{30}{x-6}$$

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**Q9.** $\dfrac{x^2}{(x^2+a)(x^2+b)(x^2+c)}$

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**Sol:** Replace $x^2$ by $y$

$$\frac{x^2}{(x^2+a)(x^2+b)(x^2+c)}=\frac{y}{(y+a)(y+b)(y+c)}$$

Suppose $$\frac{y}{(y+a)(y+b)(y+c)}=\frac{A}{(y+a)}+\frac{B}{(y+b)}+\frac{C}{(y+c)} \rightarrow (1)$$

Multiply by $(y+a)(y+b)(y+c)$ we get

$y=A(y+b)(y+c)+B(y+a)(y+c)+C(y+a)(y+b) \rightarrow (II)$

Put $y+a=0 \Rightarrow y=-a$ in eq. (ii)

$-a=A(-a+b)(-a+c)+B(0)+C(0)$

$-a=A(b-a)(c-a)$

$$A=\frac{-a}{(b-a)(c-a)}=\frac{\not{-}a}{\not{-}(a-b)(c-a)} \Rightarrow A=\frac{a}{(a-b)(c-a)}$$

Put $y+b=0 \Rightarrow y=-b$ in eq. (ii)

$-b=A(0)+B(-b+a)(-b+c)+c(0)$

$-b=+B(a-b)(c-b)$

$$-b=-B(a-b)(b-c) \Rightarrow B=\frac{b}{(a-b)(b-c)}$$

Put $y+c=0 \Rightarrow y=-c$ in eq. (ii)

$-c=A(0)+B(0)+C(-c+a)(-c+b)$

$-c=C(a-c)(b-c)$

$$-c=-C(c-a)(b-c) \Rightarrow C=\frac{c}{(c-a)(b-c)}$$

Eq. (i) Become

$$\frac{y}{(y+a)(y+b)(y+c)}=\frac{a}{(a-b)(c-a)(y+a)}+\frac{b}{(a-b)(b-c)(y+b)}+\frac{c}{(c-a)(b-c)(y+c)}$$

Replace $y$ by $x^2$

$$\frac{x^2}{(x^2+a)(x^2+b)(x^2+c)}=\frac{a}{(a-b)(c-a)(x^2+a)}+\frac{b}{(a-b)(b-c)(x^2+b)}+\frac{c}{(c-a)(b-c)(x^2+c)}$$

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**Q10.** $\dfrac{x+1}{(x-1)^2}$

**Sol:** Suppose $\dfrac{x+1}{(x-1)^2}=\dfrac{A}{(x-1)}+\dfrac{B}{(x-1)^2}\rightarrow(i)$

Multiply by $(x-1)^2$; we get

$x+1=A(x-1)+B\rightarrow(ii)$

Put $x-1=0\Rightarrow x=1$ in eq. (ii)

$1+1=A(1-1)+B\Rightarrow 2=A(0)+B\Rightarrow B=2$

Rearrange (ii)

$x+1=Ax-A+B$

Comparing co-efficient

$x$; $\boxed{A=1}$

Eq. (I) Become

Hence $\dfrac{x+1}{(x-1)^2}=\dfrac{1}{(x-1)}+\dfrac{2}{(x-1)^2}$

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**Q11.** $\dfrac{x^2+x}{(x^2-1)^2}$

**Sol:** $\dfrac{x^2+x}{(x^2-1)^2}=\dfrac{x^2+x}{((x-1)(x+1))^2}\qquad\because a^2-b^2=(a-b)(a+b)$

Let $\dfrac{x^2+x}{(x-1)^2(x+1)^2}=\dfrac{A}{x-1}+\dfrac{B}{(x-1)^2}+\dfrac{C}{(x+1)}+\dfrac{D}{(x+1)^2}\rightarrow(i)$

Multiply by $(x-1)^2(x+1)^2$, we get

$x^2+x=A(x-1)(x+1)^2+B(x+1)^2+C(x+1)(x-1)^2+D(x-1)^2\rightarrow(ii)$

Put $x-1=0\Rightarrow x=1$ in eq. (ii)

$(1)^2+(1)=A(1-1)(1+1)^2+B(1+1)^2+C(1+1)(1-1)^2+D(1-1)^2$

$2=A(0)+B(2)^2+C(0)+D(0)$

$2=4B\Rightarrow B=\dfrac{1}{2}$

Put $x+1=0\Rightarrow x=-1$ in Eq. (ii)

$(-1)^2+(-1)=A(0)+B(0)+C(0)+D(-1-1)^2$

$1-1=D(-2)^2\Rightarrow 0=4D\Rightarrow\boxed{D=0}$

Re-arrange (II)

$x^2+x=A(x-1)(x^2+1+2x)+B(x^2+1+2x)+C(x+1)(x^2+1-2x)+D(x^2+1-2x)$

$x^2+x=A(x^3+x+2x^2-x^2-1-2x)+B(x^2+1+2x)+C(x^3+x-2x^2+x^2+1-2x)+D(x^2+1-2x)$

$x^2+x=A(x^3+x^2-x-1)+B(x^2+1+2x)+C(x^3-x^2-x+1)+D(x^2+1-2x)$

$x^2+x=(A+C)x^3+(A+B-C+D)x^2+(-A+2B-C-2D)x+(-A+B+C+D)x^0$

Comparing co-efficient

$x^3$; $\qquad 0=A+C\rightarrow(iii)$

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$x^2$: $1 = A + B - C + D \rightarrow (iv)$

$x$: $1 = -A + 2B - C - 2D \rightarrow (v)$

$x^0$: $0 = -A + B + C + D \rightarrow (vi)$

Put values $B = \frac{1}{2}, D = 0$ in eq. (iv)

$$1 = A + \frac{1}{2} - C + 0$$

$$1 - \frac{1}{2} = A - C$$

$$\frac{1}{2} = A - C \rightarrow (vii)$$

Adding eq. (iii) and (vii)

$$0 = A + \not{C}$$

$$\frac{1}{2} = A - \not{C}$$

$$\frac{1}{2} = 2A$$

$$\boxed{A = \frac{1}{4}}$$

Put $A = \frac{1}{4}$ in eq. (iii)

$$\frac{1}{4} + C = 0 \Rightarrow \boxed{C = -\frac{1}{4}}$$

Put values A, B, C and D in eq. (i)

$$\frac{x^2 + x}{(x^2 - 1)^2} = \frac{1}{4(x-1)} + \frac{1}{2(x-1)^2} - \frac{1}{4(x+1)} + \frac{0}{(x+1)^2} = \frac{1}{4(x-1)} + \frac{1}{2(x-1)^2} - \frac{1}{4(x+1)}$$

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**Q12.** $\dfrac{3x^2 + 4x - 5}{(x-1)^3}$

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**Sol:** Suppose

$$\frac{3x^2 + 4x - 5}{(x-1)^3} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} \rightarrow (i)$$

Multiply by $(x-1)^3$, we get

$$3x^2 + 4x - 5 = A(x-1)^2 + B(x-1) + C \rightarrow (ii)$$

Put $x - 1 = 0 \Rightarrow x = 1$ in eq. $(ii)$

$$3(1)^2 + 4(1) - 5 = A(1-1)^2 + B(1-1) + C$$

$$3 + 4 - 5 = A(0) + B(0) + C \Rightarrow \boxed{2 = C}$$

Re-Arrange eq. (ii)

$$3x^2 + 4x - 5 = A(x^2 + 1 - 2x) + B(x-1) + C$$

$$3x^2 + 4x - 5 = Ax^2 + A - 2Ax + Bx - B + C$$

$$3x^2 + 4x - 5 = Ax^2 + (-2A + B)x + (A - B + C)x^0$$

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Comparing co-efficient

$x^2$; $\boxed{3 = A}$

$x$; $4 = -2A + B \rightarrow (iii)$

Put $A = 3$ in eq. (iii)

$4 = -2(3) + B \Rightarrow B = 4 + 6 = 10 \Rightarrow \boxed{B = 10}$

Put values A, B and C in eq. (i)

Hence $\dfrac{3x^2 + 4x - 5}{(x-1)^3} = \dfrac{3}{x-1} + \dfrac{10}{(x-1)^2} + \dfrac{2}{(x-1)^3}$

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**Q13.** $\dfrac{1}{x(x+1)^3}$

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**Sol:** Suppose

$$\frac{1}{x(x+1)^3} = \frac{A}{x} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3} \rightarrow (i)$$

'X' by both sides $x(x+1)^3$ we get

$$1 = A(x+1)^3 + B(x)(x+1)^2 + C(x)(x+1) + D(x) \rightarrow (ii)$$

Put $x = 0$ in eq. (ii)

$1 = A(0+1)^3 + 0 + 0 + 0 \Rightarrow \boxed{A = 1}$

Put $x + 1 = 0 \Rightarrow x = -1$ in eq. (ii)

$1 = A(-1+1)^3 + B(-1)(-1+1)^2 + C(-1)(-1+1) + D(-1)$

$1 = A(0) + B(0) + C(0) + D(-1) \Rightarrow \boxed{D = -1}$

Re-arrange eq. (ii)

$1 = A(x^3 + 3x^2 + 3x + 1) + Bx(x^2 + 1 + 2x) + C(x^2 + x) + Dx$

$1 = Ax^3 + 3Ax^2 + 3Ax + A + Bx^3 + Bx + 2Bx^2 + Cx^2 + Cx + Dx$

Comparing co-efficients

$x^3$; $0 = A + B \rightarrow (iii)$

Put $A = 1$ in eq. (iii)

$0 = 1 + B \Rightarrow \boxed{B = -1}$

$x^2$;

$3A + 2B + C = 0 \rightarrow (iv)$

Put values $A$ & $B$ in eq. (iv)

$3(1) + 2(-1) + C = 0 \Rightarrow 3 - 2 + C = 0 \Rightarrow \boxed{C = -1}$

Put values A, B, C and D in eq. (i)

Hence $\dfrac{1}{x(x+1)^3} = \dfrac{1}{x} - \dfrac{1}{x+1} - \dfrac{1}{(x+1)^2} - \dfrac{1}{(x+1)^3}$

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**Q14.** $\dfrac{4x^2 - 3x + 1}{(x+1)(x-1)^2}$

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**Sol:** Suppose

$$\frac{4x^2 - 3x + 1}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2} \rightarrow (i)$$

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Multiply by $(x+1)(x-1)^2$, we get

$4x^2-3x+1=A(x-1)^2+B(x+1)(x-1)+C(x+1)\rightarrow(ii)$

Put $x+1=0\Rightarrow x=-1$ in eq. (ii)

$4(-1)^2-3(-1)+1=A(-1-1)^2+B(0)+C(0)$

$4+3+1=4A\Rightarrow 8=4A\Rightarrow \boxed{A=2}$

Put $x-1=0\Rightarrow x=1$ in eq. (ii)

$4(1)^2-3(1)+1=A(1-1)^2+B(1+1)(1-1)+C(1+1)$

$4-3+1=A(0)+B(0)+C(2)$

$2=2C\Rightarrow \boxed{C=1}$

Re-arrange

$4x^2-3x+1=A(x^2+1-2x)+B(x^2-1)+C(x+1)$

Comparing co-efficient

$x^2$: $4=A+B\rightarrow(iii)$

$4=2+B\Rightarrow 4-2=B\Rightarrow \boxed{B=2}$

Eq. (i) Become

Hence $\dfrac{4x^2-3x+1}{(x+1)(x-1)^2}=\dfrac{2}{x+1}+\dfrac{2}{x-1}+\dfrac{1}{(x-1)^2}$

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**Q15.** $\dfrac{12x^2-48}{(x-2)^2(x+2)^2}$

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**Sol:** Suppose

$$\frac{12x^2-48}{(x-2)^2(x+2)^2}=\frac{A}{(x-2)}+\frac{B}{(x-2)^2}+\frac{C}{(x+2)}+\frac{D}{(x+2)^2}\rightarrow(i)$$

Multiply by $(x-2)^2(x+2)^2$ both sides we get

$12x^2-48=A(x-2)(x+2)^2+B(x+2)^2+C(x+2)(x-2)^2+D(x-2)^2\rightarrow(ii)$

Put $x-2=0\Rightarrow x=2$ in eq. (ii)

$12(2)^2-48=A(0)+B(2+2)^2+C(0)+D(0)$

$12(4)-48=B(4)^2$

$48-48=16B\Rightarrow \boxed{B=0}$

Put $x+2=0\Rightarrow x=-2$ in eq. (ii)

$12(-2)^2-48=A(0)+B(0)+C(0)+D(-2-2)^2$

$48-48=16D\Rightarrow \boxed{D=0}$

Rearrange eq. (ii)

$12x^2-48=A(x-2)(x^2+4x+4)+B(x^2+4x+4)$

$+C(x+2)(x^2-4x+4)+D(x^2-4x+4)$

$12x^2-48=A(x^3+4x^2+4x-2x^2-8x-8)+B(x^2+4x+4)$

$+C(x^3-4x^2+4x+2x^2-8x+8)+D(x^2-4x+4)$

$12x^2-48=A(x^3+2x^2-4x-8)+B(x^2+4x+4)$

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$+C\left(x^3-2x^2-4x+8\right)+D\left(x^2-4x+4\right)$

Comparing co-efficient

$x^3$: $\quad A+C=0\rightarrow(iii)$

$x^2$: $\quad 12=2A+B-2C+D\rightarrow(IV)$

$x$: $\quad 0=-4A+4B-4C-4D\rightarrow(V)$

$x^0$: $\quad -48=-8A+4B+8C+4D\rightarrow(VI)$

Put values B and D in eq. (v) and (vi)

$0=-4A+4(0)-4C-4(0)$

$-48=-8A+4(0)+8C+4(0)$

$0=-4A-4C\rightarrow(vii)$

$-48=-8A+8C\rightarrow(viii)$

Eq. (vii) X by 2 and add in eq. (vii)

$$0=-8A-\cancel{8C}$$

$$\underline{-48=-8A+\cancel{8C}}$$

$$-48=-16A \quad \Rightarrow \boxed{A=3}$$

Put $A=3$ in Eq. (iii)

$A+C=0$

$3+C=0\Rightarrow \boxed{C=-3}$

Put values A, B, C and D in eq. (i)

Hence $\dfrac{12x^2-48}{(x-2)^2(x+2)^2}=\dfrac{3}{(x-2)}+\dfrac{0}{(x-2)^2}+\dfrac{-3}{(x+2)}+\dfrac{0}{(x+2)^2}=\dfrac{3}{(x-2)}-\dfrac{3}{(x+2)}$

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